

Title: University Library Data Warehouse – Backup and Recovery Plan

Objective:

Ensure business continuity and data recovery in case of system failure, corruption, or disaster.

Backup Strategy:

Backup Type	Frequency	Storage Location	Description
Full Backup	Weekly	Local server & cloud	Complete database snapshot, including fact and dimension tables
Incremental Backup	Daily	Cloud	Only changes since last backup, reduces storage and restores faster
ETL & Scripts Backup	Weekly	Cloud repository	Includes ETL scripts, transformation rules, and staging table structures
Dashboards Backup	Weekly	Cloud & local	Dashboard files (.pbix, Excel, Tableau)

Recovery Procedure:

1. Identify the failure type: hardware, software, human error, or cyberattack.
2. Restore the most recent **full backup** from secure storage.
3. Apply **incremental backups** to bring data up-to-date.
4. Validate restored data using **checksum and sample queries**.
5. Notify stakeholders and **resume normal operations**.

Disaster Recovery Plan:

- Critical operations can resume using **replicated cloud instance** within 24 hours.
- On-site servers: maintain **hot standby** for high-priority services.
- Document lessons learned after recovery to improve future backups.

Testing & Maintenance:

- Backup restoration is **tested monthly**.
- Backup logs are reviewed weekly.
- Backup files are **encrypted and access-controlled**.